

A Juicy Haptics Enhanced Keyboard for a Touch-Typing Course for Learning Disabled Children

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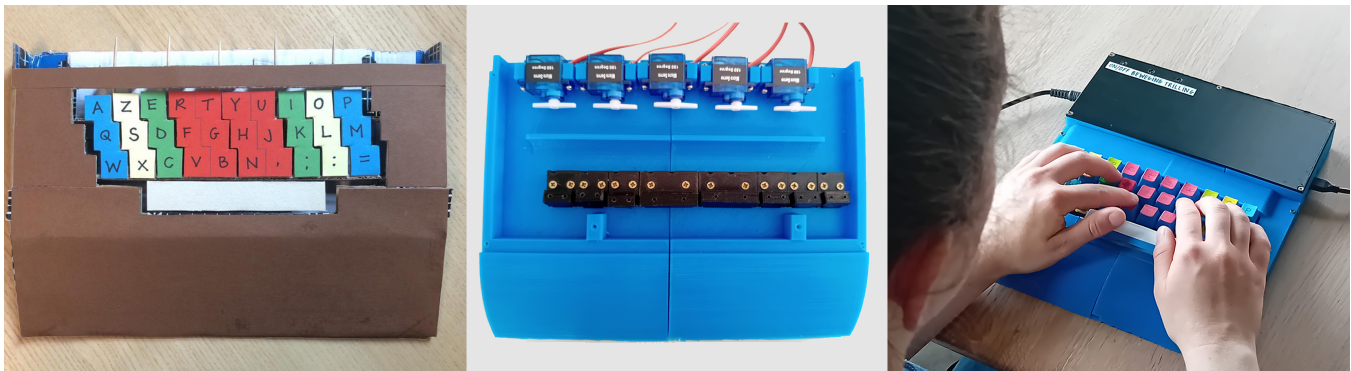


Figure 1: The evolution of the juicy haptics-enhanced keyboard, from paper prototype to final prototype, to its application in a typical use case scenario.

Abstract

Juiciness, defined as “a phenomenon emerging from coherent game mechanics and visuals, providing confirmatory, explicit, and ambient feedback,” has been widely discussed in game development, and recent research. However, most research focuses on the audiovisual aspects, neglecting the haptic domain. This paper explores “juicy haptics” by combining principles of tangible interaction and juiciness. We present a case study on designing an enhanced keyboard for a touch-typing course for children with learning disabilities and motor disorders. Two touch-typing experts collaborated with us during an iterative design process. A paper prototype, informed by juicy haptic literature and preliminary interviews, was created, eventually leading to a final keyboard prototype with two haptic

feedback mechanisms: vibration and movement. A preliminary evaluation involving non-disabled adults suggests that juicy haptics can provide a playful and pleasant learning experience for touch-typing.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; **Keyboards**; **Human computer interaction (HCI)**; *Text input*; User studies; Usability testing; *Accessibility systems and tools*.

Keywords

Children , Development / Implementation , DIY / Maker / Prototyping / Toolkits , Education , Play / Games , Tactile / Haptic Interaction , Tangible Interaction , Technology , Usability / User Experience , User Studies

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TEI '25, March 04–07, 2025, Bordeaux / Talence, France

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ACM ISBN 979-8-4007-1197-8/25/03

<https://doi.org/10.1145/3689050.3705980>

ACM Reference Format:

Zita Willems, Louise Cuypers, Ingmar Ivo P. Malfait, and Luc Geurts. 2025. A Juicy Haptics Enhanced Keyboard for a Touch-Typing Course for Learning Disabled Children. In *Nineteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '25)*, March 04–07, 2025, Bordeaux / Talence, France. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3689050.3705980>

1 Introduction

Gamification refers to “the application of game dynamics, mechanics, and frameworks into non-game settings” [30]. In education, it is used to increase student engagement and optimize learning [42]. However, loss of performance and undesired behavior of the gamification interfaces seem to be frequent occurrences [4]. Bogost, argues that these problems arise because the gamification elements are tacked on instead of fully incorporated into the interface [6]. They reward the student after the boring, educational part, rather than enhance the learning itself.

Juiciness might offer a solution as it is meant to enhance the experience of interacting with the interface itself, rather than offering a reward after completing the interaction. In gaming, juiciness describes a particular type of game feel, achieved by abundant effects [13]. Previous research has confirmed juiciness’ positive impact on user engagement and learner motivation [13, 14, 18, 41]. However, current juiciness research predominantly focuses on audiovisual modalities, neglecting the haptic and tangible experience [41]. Nevertheless, Singhal and Schneider indicates that incorporating juicy haptics in games can significantly improve engagement and immersion [41].

This research aims to deepen the understanding of “juicy haptics” through a specific case study, exploring its application in education to enhance learning. The study involves the creation of an artifact, a juicy haptics enhanced keyboard designed to support the learning of touch-typing for learning disabled children. It was developed alongside teaching experts from a touch-typing course for children with learning disabilities and motor disorders, who often learn touch-typing to address their writing difficulties [39, 48].

We argue that this target group is particularly well-suited for the study, as an adapted learning approach incorporating juicy haptics can address the unique challenges that they face. And that, touch-typing itself lends well to the different modal avenues of tangible interaction. The sensation of typing includes feeling the key, putting pressure and key travel, and feedback upon a successful press, which can be enriched by various tangible interaction modalities.

Modalities used in previous different tangible interaction interfaces including but not limited to: vibrational feedback [23], mechanical pushback and vertical movements [8], shape changing keys [37], and soft, squeezable textures [45]. Similarly, completing words, sentences and exercises also allow for tangible interaction enrichment by using e.g. ambient lighting feedback [9, 32] or ambient haptic feedback [21]. The research question is as follows: How can juicy haptics be implemented in a keyboard to facilitate the learning of touch-typing for children with learning disabilities and motor disorders?

2 Literature review

2.1 Feedback

Feedback refers to “the return of information about the result of a process or activity” [35]. It represents a product’s reaction to a user’s action and can be auditory, visual, haptic, or a combination thereof. For an interaction to be intuitive, there must be a natural coupling between action and reaction, encompassing six aspects: time, location, direction, dynamics, modality and expression [49].

In keyboards, feedback can be categorized into three types: local, partial, and global [29]. Local feedback means providing feedback only on the key that is pressed, whereas global feedback involves feedback on all keys. Partial feedback falls in between, offering feedback on a subset of keys, such as those on the left or right side of the keyboard. According to Kim and Tan, local feedback leads to better typing performance than global feedback, as it helps users identify which key was pressed [22]. However, the performance gain of local feedback must be weighed against the increased complexity and cost associated with its implementation.

2.2 Juiciness

Juiciness can be defined as abundant, continuous, and positive feedback [41]. In games, it refers to when a player’s action triggers multiple non-functional reactions [13]. The framework by Hicks et al. states that juiciness in games comprises three main components: game characteristics, game status, and immediate feedback [12]. Juicy design, according to Hicks et al., suggests that a large amount of audiovisual feedback enhances the player experience [12]. The positive effect of juiciness emerges from the combination of multiple elements that contribute to a more engaging and satisfying player experience [13]. Finally, Parker describes juiciness as a visceral feeling that provides the user with a sense of satisfaction [33].

While existing research focuses mainly on juiciness in games, we argue that this concept can be applied to other interactive media. As juicy design involves providing continuous and abundant feedback through multiple channels [13]. However, it is important that users can turn off juicy elements if they are undesirable or too overwhelming [12, 13, 41].

2.3 Haptics

Haptics relates to touch and movement, involving the mechanical interactions associated with these sensations. Haptic technology is bi-directional, encompassing both action (interaction) and reaction (haptic feedback) [16]. Examples include mobile phones simulating a click sensation when typing and game controllers vibrating when the player is hit.

The use of haptics promotes playfulness and lightweight interaction [15]. Haptic feedback can result in reduced cognitive load compared to auditory or visual feedback, leading to quicker perception [10, 31, 40]. The intensity should be carefully selected, as higher intensity can improve typing speed and accuracy, while auditory feedback intensity does not affect typing performance [29]. However, haptics, as described by Singhal and Schneider, is a multi-modal experience, linking visual, auditory, and haptic feedback [41].

Juicy haptics emerge when juiciness and haptics intersect, defined by Singhal and Schneider as “abundant, positive haptic feedback aimed at enhancing user experience in games and other interactive media [41].” This study on haptic embellishments as juicy haptics identified ten design principles and conducted two empirical studies where users interacted with visuo-haptic content on their phones. In the first study, participants compared and ranked animations with different haptic feedback conditions to evaluate

the effectiveness of each principle. In the second study, participants played six variants of a Breakout-style game with varying levels of visual and haptic juice to assess the impact of juicy haptics in games. Findings indicated that juicy haptics can significantly enhance player experience, particularly in enjoyment, aesthetic appeal, immersion, and meaning.

In this work a prototype of an interactive keyboard is designed and implemented that includes principles of juicy haptics, thereby illustrating how these principles can be translated into concrete designs. The specific use case is a keyboard that can be used in a touch-typing course for learning disabled children.

3 Case study

3.1 Envisioned end users

Children with learning disabilities and motor disorders often use a computer to overcome difficulties with writing and spelling [17, 39, 48]. Touch-typing, where each finger is linked to specific letters, is a valuable skill that can help them overcome slow handwriting [48]. Research indicates that students with learning disabilities who learn touch-typing can achieve typing speeds comparable to their peers [47]. However, their learning process is often more challenging, leading to loss of concentration and frustration. This case study aims to make this process more enjoyable, to increase motivation and improve learning outcomes.

This research was conducted in collaboration with a special needs primary school in Leuven called Windekind [50], which has developed its own learning method called TypWind. This multi-sensory approach makes associations between letters, colors and figures [46]. Each finger is linked to a specific color and letter group (e.g., the left pinky finger types A, Q, and W) and each letter is linked to a Dutch word and corresponding figure. After learning these associations, the children do typing exercises, where it is important to return to the starting position on the home row (index fingers on J and F) before typing the next letter. Despite the effectiveness of TypWind, the repetitive nature of the typing exercises causes loss of concentration and motivation.

3.2 Requirements

3.2.1 Technical requirements. The keyboard should follow the Belgian AZERTY layout, as used by the children, and conform to standard dimensions with a key distance between 17mm and 19mm for optimal typing performance and usability [34]. It should also incorporate elements of juicy haptics, providing abundant feedback through multiple channels. During an initial interview, experts of TypWind emphasized the importance of customization. The functionalities of the enhanced keyboard should be toggleable to accommodate each child's preferences [25]. Lastly, the keyboard should be ergonomically designed to avoid wrist discomfort.

3.2.2 User requirements. Children with learning disabilities and motor disorders require a different learning method than their peers, imposing specific design considerations [7]. A first requirement is that no time limit is imposed while learning to touch-type. Most typing programs emphasize speed, creating pressure for children with learning disabilities and motor disorders. The focus is on correctness without time pressure [44]. Furthermore, negative

feedback should be avoided, highlighting when something goes well, e.g. typing a correct letter [44]. The experts also emphasize this: "negative feedback is demotivating and typing speed is not important [25, 46]." Finally, the keyboard should not hinder the learning process or introduce excessive dependencies onto the children. Using technologies like gloves with sensors and actuators, glasses, and wristbands could obstruct the classroom usage of the enhanced keyboard. All mentioned requirements were imposed in consultation with Windekind to ensure the design meets the specific needs of the target group.

4 Methodology

With approval from the Social and Societal Ethics Committee of KU Leuven, an iterative design approach was used, involving experts of TypWind in every phase. The first phase explored how to integrate juiciness and haptics into a keyboard. A low-fidelity paper prototype was evaluated by two experts of TypWind. In the second phase, a high-fidelity technical prototype was evaluated by the same experts. The final phase involved a preliminary evaluation of the prototype with non-disabled adults, as ethical concerns prevented testing with the intended end users. Each evaluation consisted of one or more typing exercises, followed by a semi-structured interview.

The low-fidelity prototype was evaluated on March 28, 2024 at Windekind by two experts of TypWind. The observation involved one exercise, with key movement simulated using the paper prototype shown in Figure 1. We simulated vibration by explaining that the keyboard vibrates.

The high-fidelity prototype was evaluated on April 23, 2024 at Windekind by the same two experts of TypWind. The observation consisted of four exercises: no vibration or movement (1), vibration and movement as feedback (2), vibration as feedforward and movement as feedback (3) and movement as feedforward and vibration as feedback (4).

The evaluation with non-disabled adults took place over several days, at various locations. The test group included 13 participants aged 20 to 77, with six women and seven men. The observation included six exercises: no vibration or movement (1), vibration as feedback (2), movement as feedback (3), vibration and movement as feedback (4), vibration as feedforward and movement as feedback (5) and lastly movement as feedforward and vibration as feedback (6).

5 Design and Development

5.1 Conceptual design

The implementation of juiciness in hardware is still at an early stage. Existing techniques for implementing juiciness in games cannot be directly applied to a keyboard but are used as inspiration [13]. As mentioned in the literature review, one study has investigated juicy haptics, but it only explored vibration. Since there are currently no keyboards available that implement abundant haptic feedback, it was decided to design a keyboard from scratch. We only implement the three letter rows and the space bar, as this simplifies the further design process of the keyboard. Moreover, TypWind focuses on these keys [25, 46].

When implementing juiciness, it is important to provide continuous and immediate feedback [41]. Hence the choice to provide feedback after each typed letter. Since the user requirements dictate that negative feedback is not allowed, feedback is only given after typing a correct letter. This feedback is immediate, respecting the aspect ‘time’ of natural coupling [49].

In juicy design, abundant feedback, which can be visual, auditory, haptic, or a combination [12], is crucial. This research focuses on haptic feedback due to its benefits, including improved typing performance [22], playfulness, and lightweight interaction [15]. Haptic feedback also reduces cognitive load and is perceived faster than auditory or visual feedback [10, 40]. The keyboard implements several forms of haptic feedback, while audiovisuals are minimized. Firstly, the keyboard provides vibration. This is inspired by screen shakes in games and existing implementations of haptic feedback, such as vibrating game controllers. In addition, the keys can move upwards. The idea is to exaggerate the upward movement that the finger makes after pressing a key. This exaggerated response follows from the component ‘game state’ of the framework of juicy design of games [12]. Since the feedback follows the movement of the finger, the aspect ‘direction’ of natural coupling is respected [49].

According to Interhaptics, vibration belongs to tactile technology and movement to kinesthetic technology. By making this choice, two different feedback mechanisms can be tested [16]. These two feedback mechanisms can occur simultaneously, separately or not at all. Thus, the keyboard is adaptable to the various needs of the children, a requirement imposed by the experts of TypWind [25].

As mentioned in the literature review, local feedback, where each key gives feedback individually, results in a higher typing speed and a lower error rate [21, 22]. However, the size of hardware components and the requirement to maintain standard keyboard dimensions must be considered. It was therefore decided to provide feedback per letter group. The keyboard is divided into the same colored letter groups the children learn in TypWind.

Upon completing a typing exercise, the keyboard makes a wave motion, with each letter group providing feedback sequentially. The exuberant feedback emphasizes and rewards the achievement, following the component ‘game characteristics’ of the framework of juicy design of games [12]. The wave is delivered through either vibration or movement depending on which modality is active, with movement preferred when both are active. Finally, texture is added to the eight keys of the starting position. This tactile enhancement aids children in easily locating and keeping their fingers on these keys while typing, facilitating better finger placement and consistency.

5.2 Technical design

A possible way to implement juicy haptics in a keyboard is by incorporating multiple haptic feedback mechanisms to provide continuous and immediate feedback. This feedback is given after each correctly typed letter, thereby emphasizing progress. The first mechanism is vibration, inspired by screen shakes in games and existing implementations of haptic feedback like vibrating game controllers. The second mechanism is movement, where the upward movement of a finger after pressing a key is exaggerated. And at

the end of a typing exercise, the keyboard makes a wave motion to emphasize completion. Through the multi-modal coherence of vibration and movement as feedback, alongside the tactile feel on the home row, the result is a juicy haptics enhanced keyboard, according to the definitions in section 2.

5.2.1 Hardware. The technical prototype features keys to enable the basic operation of the keyboard, as well as additional components to provide interaction with the user. Clicky mechanical switches are used for their tactile feedback, audible click and low profile [5, 11, 19, 20, 24], ensuring the keyboard maintains an appropriate height.

As detailed in subsection 5.1, the enhanced keyboard can provide feedback through both movement and vibration. The upward movement of the letter groups is achieved by servo motors. Additionally, vibration motors offer a subtle vibration for each letter group, with the space bar featuring two motors to ensure noticeable vibration in both thumbs. Lastly, texture is applied on the starting keys of the home row using felt. This creates a different feel compared to the other keys, making it easier for the user to find the starting position. The sketch of the design is shown in Figure 2.

5.2.2 Software. A Raspberry Pi (RPI) model 4B is used to control the servo and vibration motors and to read the pressed keys [38]. The application is developed in Python, because of the multiple libraries available for the hardware components [1–3, 43]. A graphical user interface (GUI) is implemented to display the letters to be typed, which is shown on an external screen [36]. The block diagram illustrating the various components is provided in Figure 3.

5.2.3 Mechanical design. To ensure user-friendliness, the keyboard includes a casing to conceal wires and enhance usability. The casing is composed of multiple 3D printed parts. The front part of the keyboard is rounded for comfortable wrist support, improving ergonomics as required in by Typwind. The technical prototype is shown in Figure 1.

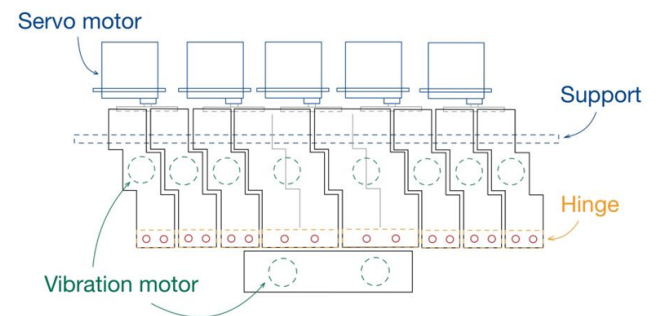


Figure 2: Top view of the keyboard design sketch. Servo motors are arranged along the top to maneuver various letter groups shown in the center, enabling vertical movement of the upper end while a hinge at the bottom maintains their level. Vibration motors are installed beneath the letter groups and the space bar at the bottom.

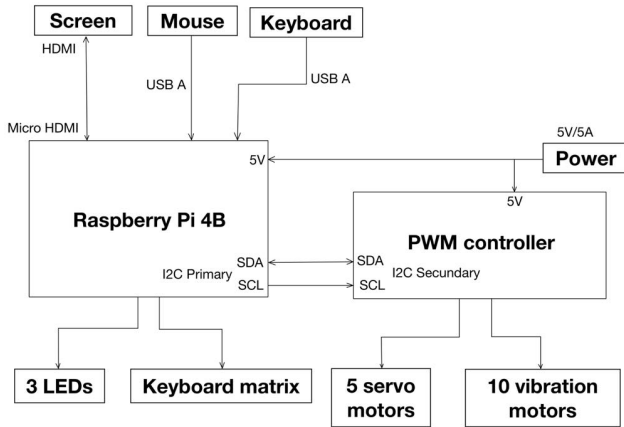


Figure 3: Block diagram of the enhanced keyboard. The Raspberry Pi processes keyboard and mouse inputs (USB) and controls the servo and vibration motors via a separate PWM controller (I²C). The GUI is displayed on the screen (HDMI).

6 Results evaluations

6.1 Results evaluation paper prototype

As previously discussed in section 4 on methodology, the design underwent multiple testing phases. The evaluation of the paper prototype, depicted in Figure 1, by two experts of TypWind yielded the following findings [26]. One expert noted that when they removed their fingers from the keyboard, they could no longer feel the feedback. They suggested that this serves as an effective way to encourage children to keep their fingers on the keyboard in order to feel the feedback. The keyboard was also described as low-stimulus, directing more attention to the letters. Both experts highlighted the compact design of the keyboard as beneficial, as it minimizes distractions for children. Furthermore, they considered continuous feedback a significant improvement over receiving feedback only at the end of a typing exercise. The absence of negative feedback was seen as motivating, and the use of colors was appreciated, as it creates a connection with the learned associations, enhancing the overall effectiveness of the keyboard. Additionally, one expert suggested incorporating feedforward, where users receive information before performing an action. Although not strictly related to juiciness, the design allows for this addition through software adjustments. Further testing will determine whether this feature enhances motivation. Lastly, one expert suggested using lowercase letters instead of uppercase letters, since the children typically use lowercase letters during class activities.

6.2 Results evaluation final prototype

As detailed in section 4, the high-fidelity prototype was tested by the same two experts of TypWind who evaluated the paper prototype [27, 28]. Both experts reported that the texture on the home row effectively indicates the starting position. They also found the tactile click sensation of the mechanical switches enjoyable and stimulating. While typing, both experts were pleasantly surprised by the vibration and movement of the keys. They noted that it

creates an enjoyable sensation, but that simultaneous vibration and movement could be overwhelming for some children. Nevertheless, the experts believe that children might find these haptic sensations enjoyable and motivating, as they spark curiosity. Both experts appreciated the addition of positive feedback and the absence of negative feedback. Moreover, they found the continuous feedback after each letter motivating, as it encourages further typing. However, their opinions on feedforward differed. One expert struggled with correctly interpreting feedforward through vibration, but found it helpful when implemented through movement. The other expert perceived feedforward as unclear and rather distracting, as they were more focused on the associations taught during TypWind. Lastly, both experts praised the keyboard's simple layout and compactness, along with its ability to customize the two feedback modalities according to user preferences. They emphasized the importance of adaptability, given that each child is unique. One expert suggested that feedforward might be more effective with a longer interval between the feedback for the previous key and the feedforward for the next key. Consequently, this time was increased for the evaluation with non-disabled adults.

6.3 Results evaluation with non-disabled adults

The evaluation with non-disabled adults offers initial insights into the user experience of the juicy haptics enhanced keyboard. The felt on the starting position was perceived as pleasant and helpful. While most participants appreciated the addition of vibration and movement, some had difficulty adjusting or found it distracting. Overall, participants reported that the combination of vibration and movement creates an enjoyable and enhanced experience, although some felt it could be overwhelming.

The majority preferred the integration of feedforward and feedback, as it reduces cognitive load during typing. Lastly, most participants enjoyed the wave motion at the end, describing it as playful and rewarding, which adds to the sense of accomplishment.

7 Discussion and future work

The evaluation with non-disabled adults suggests that the addition of vibration and movement, along with the wave motion at the end, creates an enjoyable and playful typing experience. These findings are consistent with existing literature suggesting that juicy haptics contribute to a positive player experience [41]. However, the combination of both vibration and movement can be overwhelming or distracting, underscoring the need for both modalities to be individually toggleable. This confirms that juiciness should be used in moderation [12, 13, 41]. Additionally, it emphasizes the need for customization, highlighting the importance of adaptability based on user preferences.

While the keyboard is specifically designed for children with learning disabilities and motor disorders, it is suitable for all users. Nevertheless, its benefits are particularly significant for these children, as their touch-typing learning process is more challenging, leading to loss of concentration and motivation. Lastly, it is important to note that testing with the intended end users was not possible due to ethical concerns. As a result, no definitive conclusions can be drawn regarding children with learning disabilities

and motor disorders. In the future, we hope to conduct a user evaluation with children with learning disabilities and motor disorders to assess the effect of the juicy haptics enhanced keyboard on their motivation and learning process.

8 Conclusion

This study explores how juicy haptics can be integrated into a keyboard to facilitate the learning of touch-typing for children with learning disabilities and motor disorders. An iterative design approach was adopted in collaboration with experts of TypWind. Juicy haptics are implemented through two haptic feedback mechanisms, vibration and movement, which provide continuous and immediate feedback after each correctly typed letter. Additionally, the keyboard makes a wave motion upon completing a typing exercise to highlight this achievement.

In conclusion, juicy haptics show promise in creating a pleasant and stimulating experience for children with learning disabilities and motor disorders as they learn to touch-type. Although the prototype was not tested with the intended end users, a preliminary evaluation with non-disabled adults suggests that juicy haptics, implemented as vibration and movement, can enhance the overall typing experience.

Acknowledgments

We would like to thank Windekind, particularly the experts of TypWind, for making this research possible. Thanks to their interviews, we received valuable feedback on both our paper and technical prototypes. We also extend our gratitude to the children who provided us with insight into their learning process.

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